Luke Rodel

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EDUCATION

University of Alabama — B.S. Mechanical Engineering candidate

Expected May 2027 | Robotics minor candidate | GPA: 3.96 / 4.0

ENGINEERING EXPERIENCE

Herrick Technology Laboratories

Mechanical Engineering Intern — May–August 2026

- Repaired CAD models, drawings, and bills of materials to improve consistency and support manufacturing.
- Developed thermal testing practices using physical testing and Autodesk CFD; supported mechanical packaging and environmental test preparation.
- Created Creo hole definitions for threaded inserts and automated drawing callouts.

Mercedes-Benz U.S. International

Industrial Engineering Co-op — May–August 2025

- Built Power BI tools for quality analysis and automated monthly scrap reporting, reducing preparation from roughly six hours to minutes.
- Investigated data inconsistencies, mapped scrap and chargeback workflows across departments, and proposed improvements to material traceability.
- Supported department exploration of AI and automation opportunities with the data team.

Mercedes-Benz U.S. International

Battery Series Planning Co-op — August–December 2024

- Designed an angled printer stand and helped integrate automatic label vision grading, reducing the label workflow by approximately ten seconds.
- Developed an automated data comparison tool for error-log verification; supported maintenance and production workflow improvements.

Spudnik Equipment Company

Design Engineering Intern — May–August 2024

- Redesigned a sugar beet harvester drivetrain for improved tensioning and assembly access, reducing the assembly by ten parts.
- Completed more than 100 design modifications in NX to support fabrication, tooling, and assembly.
- Worked across manufacturing processes and incorporated regulatory safety labeling into CAD models.

Mercedes-Benz U.S. International

Maintenance Engineering Co-op — January–May 2024

- Designed and fabricated support brackets for front-end alignment; reverse-engineered replacement parts in CAD.
- Designed roller storage and a mobile camera cart; supported improvement proposals and vendor bids.

Electric Motor Service Company

Electric Motor Repair Intern — June–August 2021

SELECTED PROJECTS

- Senior capstone: robotic-arm direct ink writing system and end effector for non-planar polymer lattices. In progress.
- Personal projects include local AI models and agents, web apps, computers and a home media server, Ethernet and security-camera installation, and residential construction.

SKILLS

- NX, SolidWorks, Creo, Autodesk CFD, Power BI, Power Query, MATLAB, Python, Docker, 3D printing, machining, and welding.